

MA2003B Application of Multivariate Methods in Data Science

Challenge Project Guide

Measuring Air Quality with SIMA

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Chapter 1

Guide for Handling Missing Data in Time Series (Air Quality Context)

In this challenge, you are working with real-life air quality data that contains missing values. Your goal is to handle these gaps in a way that is **accurate, transparent, and reproducible**, since the results will be shared with our industry partner SIMA which is a govermnment agency and they need to justify their decisions based on your analysis. When dealing with missing data, you should keep in mind the following steps and best practices:

1.1 Clarify Your Objective

Before choosing an imputation method, decide what the imputed data will be used for:

- **Forecasting:** If you need to predict future values, you must preserve the temporal structure of the series.
- **Trend Monitoring / Reporting:** If the task is to calculate averages (e.g., yearly PM2.5 mean), you need imputations that avoid biasing the mean.
- **Statistical Analysis:** If the data will be used in regression or other models, imputations should not distort correlations between variables.

1.2 Characterize Missingness

You should be able to clearly state the general patterns of the missing data:

- Calculate the percentage of missing values per variable and per station.
- Visualize where the gaps occur (e.g., plot missingness over time).
- Check if missingness is random or shows patterns (e.g., whole days missing due to sensor outages).

1.3 Try Multiple Imputation Methods

You should compare at least two different approaches:

- **Simple Baselines:**
  - Forward/Backward fill (for short gaps only)
  - Linear interpolation
  - Seasonal averages (using historical values from the same hour/day/season)
- **Model-Based:**
  - Kalman filtering or state-space models (common in environmental monitoring)
  - ARIMA or SARIMA-based imputation (using local dynamics)
  - Multivariate imputation (using correlations across pollutants, e.g., PM2.5 and PM10)

1.4 Evaluate Imputation Quality

Since the true missing values are unknown, use indirect evaluation:

- **Internal consistency:** Check whether the mean, variance, and seasonal patterns of the series are preserved. Compare the autocorrelation (ACF/PACF) before and after imputation.
- **External validation:** Compare with nearby stations or known events (e.g., spikes during wildfires, drops after rain).
- **Downstream performance:** Fit a forecasting model (such as SARIMA) to the imputed series and compare accuracy across different imputation methods.

1.5 Document Your Process

You must keep a clear record of:

- The percentage of data imputed.
- The methods you tested and why you chose the final one.
- The raw and imputed datasets (do not overwrite the original).

1.6 Best Practices

- Use simple methods for short gaps; reserve more advanced methods for larger gaps.
- Do not over-impute: if long blocks are missing (e.g., weeks), report them as gaps instead of filling them in.
- Ensure reproducibility: write your code so others (including our industry partner) can reproduce your results.

- Be transparent: always note which values were imputed in your reporting.

1.7 Suggested Workflow

1. Quantify and visualize missingness.
2. Apply at least one baseline and one advanced imputation method.
3. Evaluate the results using internal consistency, external validation, and forecasting performance.
4. Choose the method that balances accuracy, interpretability, and reproducibility.
5. Document everything clearly for our industry partner.